

19) Show that each conditional statement in Exercise 11 is a tautology using the fact that a conditional statement is false exactly when the hypotheses is true and the conclusion is False.

a) $(p \wedge q) \rightarrow \neg p$.

Suppose for the sake of contradiction, this is not a tautology. By the defn. of conditional, there exists an assignment of truth values of p, q such that $(p \wedge q)$ is True but $\neg p$ is False.

By defn. of AND, p is True, q is True. But this is a contradiction.

$\therefore$ It is a tautology

b) $p \rightarrow (p \vee q)$

Suppose for the sake of contradiction, this is not a tautology. $\therefore$ There exists an assignment of truth values to p, q such that $p \rightarrow (p \vee q)$ is False.

By defn. of conditional, p is True and $(p \vee q)$ is False. By defn. of OR, both p, q are False. But this is a contradiction.

$\therefore$ It is a tautology

c) $\neg p \rightarrow (p \rightarrow q)$

Suppose for the sake of contradiction, this is not a tautology. $\therefore$ There exists an assignment of truth values to p, q such that $\neg p \rightarrow (p \rightarrow q)$ is False.

By defn. of conditional, $\neg p$ is True and $(p \rightarrow q)$ is False. By defn. of conditional, p is True and q is False. By defn. of negation, $\neg p$ is False. But this is a contradiction.

$\therefore$ It is a tautology

d) $(p \wedge q) \rightarrow (p \rightarrow \neg q)$

Suppose for the sake of contradiction, this is not a tautology. $\therefore$ There exists an assignment of truth values to p, q such that $(p \wedge q) \rightarrow (p \rightarrow \neg q)$ is False.

By defn. of conditional, $(p \wedge q)$ is True and $(p \rightarrow \neg q)$ is False. By defn. of AND, p, q are both True. But by defn. of ~~True~~ conditional, p is True and q is False. The truth values of q contradict each other.

$\therefore$ It is a tautology